HAVOK Burst Warning and mHAVOK Model Selection

Final project results for Hindmarsh-Rose and Lorenz

Hindmarsh-Rose

- Question: does the HAVOK forcing signal switch on before burst onset?
- Result: **yes for onset warning**, with strong recall across practical lead windows.

Lorenz

- Question: how sensitive is mHAVOK model ranking to rank, delays, and observable choice?
- Result: **very sensitive**. The winner changes with the objective and channel set.

HR BURSTS

BEST HR RECALL @ 20 TU

BEST LORENZ COMBO

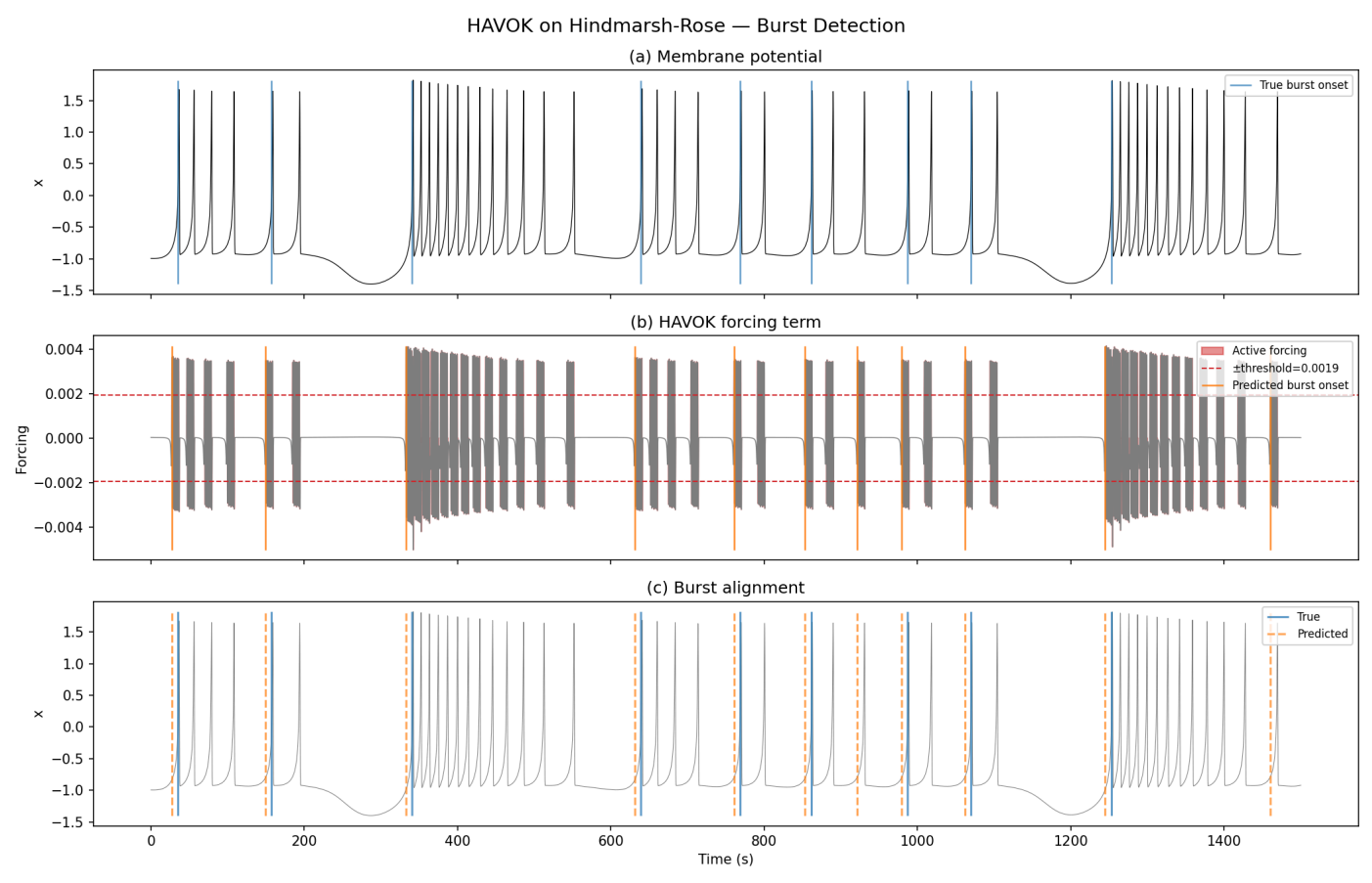
Hindmarsh-Rose Bursting Regime

Setup summary

- Total simulated time: **49049.9** time units
- Bursts detected: **216**
- Mean inter-burst interval: **227.8**
- IBI standard deviation: **81.4**
- Coefficient of variation: **0.36**

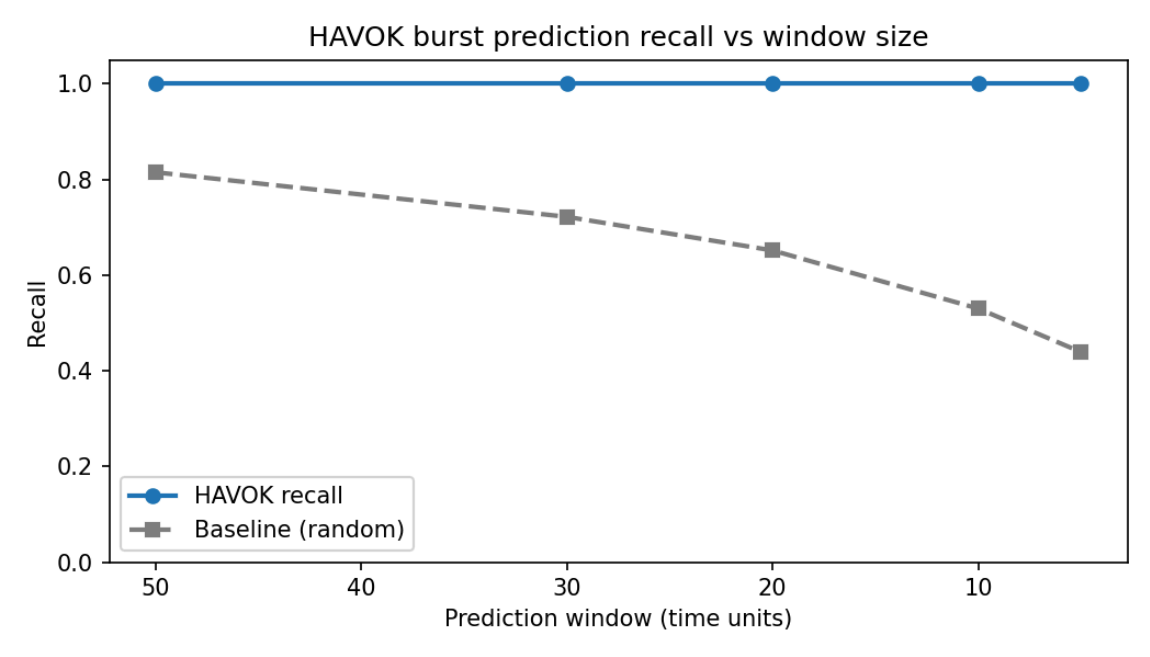
Why this matters

- The target is **burst onset**, not generic large amplitude activity.
- The regime is structured



Burst detection trace from the HR simulation.

HR Forcing Works as an Early-Warning Signal



Recall remains perfect across 50, 30, 20, and 10 time-unit windows.

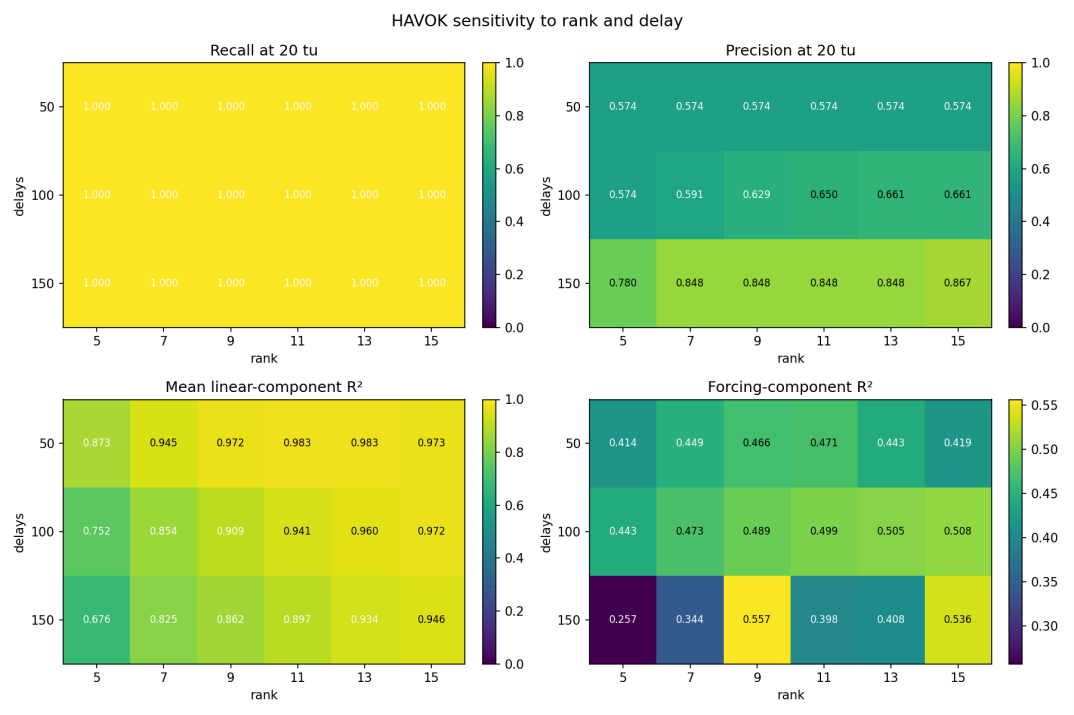
Core result

- Forcing threshold: **0.001931**
- Active fraction: **0.203**
- Predicted burst onsets: **309**
- Mean lead time: **8.0** time units

Interpretation

- Every true burst is preceded by forcing activation over practical lead windows.
- Precision is **0.699** at 50, 30, 20, and 10 time units, so the signal is selective enough to be useful even though it is

HR Sensitivity to Rank, Delays, and Channel



Best x-channel setting on the tested grid: `delays = 150`, `rank = 15`.

Best tested settings

Channel	Delays	Rank	Recall	Precision	F1	Mean linear R^2
x	150	15	1.000	0.867	0.929	0.946
y	150	15	1.000	0.867	0.929	0.938
z	150	15	1.000	0.848	0.918	0.972

Takeaway

- The warning result is not coming from one fragile setting.
- **x and y** are best for event detection on the

HR Caveat: Warning Success Is Not Full-State Success

Strong warning result

- Recall at 20 time units: **1.000**
- Precision at 20 time units: **0.699**
- Lead time: **8.0** time units on average

Weak generative result

- Forcing-mode R^2: **0.5128**
- Train reconstruction R^2_rec: **-1935.3039**
- Test reconstruction R^2_rec: **-518.8316**

Statistical mismatch in free-running behavior

- True IBI mean: **227.8**
- Predicted IBI mean: **159.0**
- KS statistic: **0.3191**, p-value **0.0000**
- True exceedances above 1.5: **6360**
- Reconstructed exceedances above 1.5: **145401**

Claim we can defend

- HAVOK is useful here as an **event-warning detector**

Lorenz: The Best Model Depends on the Objective

Default `x+z` setup on the tested rank-delay grid

Objective	Delays	Rank	Median abs gap (s)	Recall	Precision	F1	Mean linear R^2
Best event F1	150	11	0.0480	0.6429	0.2535	0.3636	0.9382
Best alignment	150	13	0.0275	0.6071	0.2208	0.3238	0.9493

What changed

- The notebook sweep sorts alignment and event metrics separately.
- Those two objectives **do not pick the same model**.

Presentation consequence

- The baseline-versus-tuned overlay in the notebook is showing the **best alignment** model.
- It should not be described as the **best**

Lorenz Observable Choice Changes the Ranking

Channel combo	Delays	Rank	Recall	Precision	Accuracy	F1	Mean linear R^2	Chamfer (s)
x+y+z	150	5	0.7143	0.4878	0.8924	0.5797	0.7450	0.1488
x+y	150	5	0.6429	0.3273	0.8794	0.4337	0.9988	0.2095
x	150	5	0.6429	0.3214	0.8792	0.4286	0.9995	0.1937
x+z	150	11	0.6429	0.2535	0.8801	0.3636	0.9382	0.2352

Main result

- The best event-detection model on the tested grid is **x+y+z**, not the earlier x-only or x+z variants.

Why this matters

- Adding `y` changes the ranking materially.
- The strongest event model does **not** maximize mean linear R^2.

Lorenz Diagnostics: Most Modes Fit Well, the Last One Does Not

Tuned component fit summary

Component block	R^2
v1 through v7	1.0000
v8	0.9991
v9	0.9989
v10	0.9992
v11	0.9986
v12	0.3957

Interpretation

- The linear latent structure is fit extremely well for most modes

Related Lorenz objective table

Observable set	Tight alignment	Balanced	Early warning
z only	winner		
x and z		winner	
x only			winner

This earlier objective-selection table and the full combo benchmark tell the same story: **the winner depends on the objective**.

Final Takeaways

Hindmarsh-Rose

- HAVOK forcing is a **strong early-warning signal** for burst onset.
- The result is robust across the tested windows and across multiple input channels.
- The current model should **not** yet be presented as a high-fidelity free-running burst surrogate.

Lorenz

- In mHAVOK, **rank, delay, objective, and observable choice all matter**.
- The best alignment model, best event-F1 model, and best full observable set are different.
- The practical lesson is to choose the model **based on the question you want to answer**.

HR BEST TESTED PRECISION

0.867

LORENZ BEST COMBO F1

0.580

MOST HONEST SUMMARY

warning yes, surrogate not yet